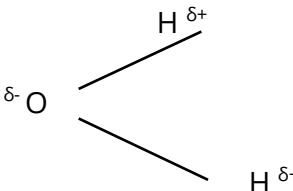


Question number	Answer	Additional guidance	Mark
5(a)	An explanation that includes the following points: - because it shows that it is {made of two layers of phospholipid / a bilayer} (1) - the size of a phospholipid is in the range 2.05 nm to 2.65 nm (1) - therefore the width of the membrane (5 nm) is within a bilayer range (of 4.1 nm to 5.3 nm) (1)	Accept it shows two layers of phosphate heads separated by a band of lipids	(3)

Question number	Answer	Additional guidance	Mark
5(b)	An explanation that includes three of the following points: - cholesterol is a non-polar molecule (1) - fatty acid tails are non-polar (1) - cholesterol will be located within the {fatty acid tails / non-polar part} of membrane (1) - OH group will be located near the phosphate heads (1)	Accept hydrophobic for non-polar in all points Accept {hydrocarbon tail / hydrocarbon rings} are non-polar	(3)

Question number	Answer	Additional guidance	Mark
5(c)(i)	A description that includes the following points: {small / weak / delta / partial} positive charge on hydrogen (1) {small / weak / delta / partial} negative charge on oxygen (1)	Accept both points from a diagram 	(2)

Question number	Answer	Additional guidance	Mark
5(c)(ii)	An explanation that includes the following points: - water is small enough to move by {osmosis / diffusion} (between the phospholipids) (1) - steroid (is non-polar so) can diffuse through the {membrane / phospholipids} (1) - glucose (is polar so) passes through {protein channels / carrier proteins / by facilitated diffusion} (1) - ions (are polar so) pass through {protein channels / carrier proteins / by facilitated diffusion / active transport} (1)	Accept description of water moving through {protein channels / aquaporins} Accept steroid is not repelled by fatty acid tails Accept movement by active transport Accept ions are repelled by fatty acid tails so cannot get through membrane	(4)